

A ternary mixed-support certificate for an Erdős–Straus shell

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Abstract

On one complete marked divisor source, three explicitly overlapping observations have a positive signed direction although each two-observation restriction is negative definite. The certificate uses only mixed gcd moments through cubic order and returns an original positive integral witness. We give the full matrix, rational trial vector, support definitions and reconstruction. The original counting metric and mixed cross terms are retained. This is a strict sufficient-test improvement, not a universal Erdős–Straus theorem.

Original states. Let $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, $R = 4a - p$, and $\gcd(pa, R) = 1$. For $a = \prod q_i^{e_i}$ retain every $c \in \{E, M\}$ and $\beta_i \in [-e_i, e_i]_{\mathbb{Z}}$. Put

$$u = \prod_i q_i^{e_i + \beta_i}, \quad g_E = 4u + 1, \quad g_M = u + a, \quad D = R / \gcd(R, g_c), \quad x = 1/D.$$

The positive rational triples are

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

They satisfy $(RY - pa)(RZ - pa) = p^2a^2$ and hence have reciprocal sum $4/p$. Reduction modulo R shows that both nonfirst reduced denominators equal D . Thus a hit is exactly $D = 1$; every failure has odd $D \geq 3$. These marked Type I/II coordinates are inherited [1].

Mixed observations, not independent sources. For original counting space $\mathbb{R}^\Omega$ and specified functions $f_0, \dots, f_{r-1}$ on the same states, define

$$(Av)_\alpha = \sum_i v_i f_i(\alpha), \quad K = A^*A, \quad J_{ij} = \sum_\alpha (x_\alpha - 1/3) f_i(\alpha) f_j(\alpha). \quad (1)$$

These are signed localizing forms [2]. Their cross terms record actual overlap. The kernel of K is exactly $\ker A$ and lies in $\ker J$. The original norm on the coefficient quotient is v^*Kv , not an assigned identity matrix. The source’s joint Schur calculation [3, JSA12–JSA20] likewise keeps its complete cross block; its analytic measure is not identified with (1).

Theorem 1 (Positivity, count and a strict ternary test). *Let H count original states with $D = 1$. Then $n_+(J) \leq H$. If $|(Av)_\alpha|^2 \leq M(v)$ on every possible successful pattern, then*

$$H \geq \left\lceil \frac{3v^*Jv}{2M(v)} \right\rceil \quad (M(v) > 0). \quad (2)$$

For a symmetric 3×3 matrix with negative diagonal entries and positive three two-coordinate principal determinants, a positive direction exists exactly when its full determinant is positive. In that case its inertia is $(1, 2, 0)$ while all two-coordinate restrictions are negative definite.

Proof. Failures contribute nonpositively to v^*Jv , while hits contribute $(2/3)|(Av)_\alpha|^2$. This proves (2). Restricting ΠA to a positive subspace of J is injective: a vector with zero successful coordinates has nonpositive form. Hence its dimension is at most H . For the ternary assertion write $J = \begin{pmatrix} -a & b^* \\ b & c \end{pmatrix}$. The invertible substitution $s = t - b^*z/a$ gives

$$(t, z)^* J(t, z) = -as^2 + z^*(C + bb^*/a)z. \quad (3)$$

The latter 2×2 matrix has negative diagonal entries by the two-coordinate conditions, and determinant $-\det J/a$. This proves all inertia claims. $\square$

Actual arithmetic certificate. Use the previously certified prime [4]

$$p = 2671180768668258904496300170662649$$

$$a = (p + 135)/4 = 2^5 431^3 457^3 7229^4, \quad R = 135.$$

Its complete original source has $2 \cdot 11 \cdot 7 \cdot 7 \cdot 9 = 9702$ states. Define supports without referring to successful-state labels:

$$\mathcal{E} = \{c = E\}, \quad \mathcal{B} = \{\beta_{457} > 0\}, \quad f_0 = 1, \quad f_1 = x\mathbf{1}_{\mathcal{E}}, \quad f_2 = x\mathbf{1}_{\mathcal{B}}. \quad (4)$$

The third predicate means $v_{457}(u) > 3$. It is an original bounded-exponent condition, not a successful-state label. Let S_k^{ij} denote the sum of $\gcd(R, g)^k$ in the four cells $(\mathbf{1}_E, \mathbf{1}_B) = (i, j)$. Exact enumeration gives

(i, j)	S_0	S_1	S_2	S_3
(0, 0)	2772	19480	435596	14842144
(0, 1)	2079	14819	332487	11329307
(1, 0)	2772	19474	435524	14841442
(1, 1)	2079	14805	368151	17899341

(5)

These moments also follow from multiplying the four actual bounded exponent polynomials modulo 135, retaining the 457-half flag. This independent calculation is included in the verifier. The full signed matrix is

$$3R^3J = \begin{pmatrix} -20121056100 & -299246400 & -256139010 \\ -299246400 & -10273776 & 3997638 \\ -256139010 & 3997638 & -6900186 \end{pmatrix}. \quad (6)$$

The two-coordinate determinants are

$$117170815341873600, \quad 73231837162654500, \quad 54909855743292,$$

whereas the full determinant is

$$799917306693954510200400 > 0. \quad (7)$$

Thus every pair of the three *specified* observations fails, and their full mixture succeeds. This does not exclude a different reparametrization that names an already combined function as one column.

An explicit rational positive trial is

$$F = \frac{-1 + 36x\mathbf{1}_E + 36x\mathbf{1}_B}{71}.$$

At $D = 1$ it takes only the values $-1/71, 35/71, 1$, so $|F| \leq 1$ there. Multiplying (6) by $(-1, 36, 36)$ gives

$$\frac{3}{2} \sum_{\alpha} (x_{\alpha} - 1/3) F_{\alpha}^2 = \frac{607462}{1890375} > 0. \quad (8)$$

Theorem 1 proves a positive integral ES witness. Deleting just the mixed entry would give the negative value $-1639754/17013375$ instead, losing the proof. The older scalar first-degree test on $(1, x)$ fails on the combined source and on each channel separately, as recalculated in the certificate [4].

An actual marked return. One reconstructed hit has

$$\begin{aligned} \beta &= (5, -2, 3, -4), \quad u = 4020447794899891321856, \quad c = E, \\ h &= 431, \quad r = 3054207776, \quad s = 507303010196865101041, \\ \kappa &= 60432156109394470448106286277616196397479. \end{aligned}$$

Directly, $a = hrs$, $u = hr^2$ and $135\kappa = pr + s$. Therefore $(a, hsk, phrk)$ is a positive integer triple with reciprocal sum $4/p$. The ordered inverse is $u = a^2/(RY - pa)$. Every other returned coordinate follows by the stated gcd normalization; none of the denominator order, prime exponents or channel is suppressed.

Original metric and all source information. The actual positive source Gram for (4) is

$$K = R^{-2} \begin{pmatrix} 176818950 & 4627665 & 3999240 \\ 4627665 & 803675 & 368151 \\ 3999240 & 368151 & 700638 \end{pmatrix}.$$

It is nonsingular: the actual rows $(M, 1), (E, 1), (M, 457^4)$ give independent coordinate readings. They recover the coefficients from the image by $c_0 = F(M, 1)$, $c_1 = (F(E, 1) - c_0)/x(E, 1)$ and $c_2 = (F(M, 457^4) - c_0)/x(M, 457^4)$. The full inverse of an arbitrary source decomposition retains its orthogonal remainder $(I - P_A)F$. It is not a claim that three observations encode all 9702 original states.

For a budget split $e = f + b$, the complete fibre over β is

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi.$$

Its size n_β is the product of these interval lengths. Duplicating source values into occurrences with weight $1/n_\beta$ is an isometry; its adjoint averages each fibre. Thus it preserves K, J and all their mixed entries. The upper-half predicate is evaluated on $\beta = \xi + \eta$, not on ξ . The tested split $f = (5, 2, 2, 4), b = (0, 1, 1, 0)$ has 44550 occurrences but original mass 9702; seven raw hit occurrences have arithmetic mass three.

Uniform mixed filters, and a necessary correction. The inherited polynomials [4]

$$\Phi_j(x) = x \prod_{\nu=1}^j \frac{(2\nu+1)x-1}{2\nu}, \quad \varepsilon_j = \frac{1}{2^j j! (2j+3)}$$

have one-sided factorial error on the odd reciprocal spectrum. Consequently, for *any* feature synthesis,

$$0 \preceq (-1)^j \{A^* \Phi_j(X)A - A^* \Pi A\} \preceq \varepsilon_j K. \quad (9)$$

This follows by evaluating on a coefficient vector and using the pointwise bound with weight $|(Av)_\alpha|^2$. Passing to the quotient with its original K metric makes A isometric. Direct sums over arbitrary growing prime-indexed feature spaces therefore retain the same operator-norm error ε_j . This bound controls mixed observations, not their unknown positive lower limit.

One must filter *before* compression. If $\bar{A}$ is isometric, $C = \bar{A}^* X \bar{A}$ and $P_A = \bar{A} \bar{A}^*$, then

$$\bar{A}^* X^2 \bar{A} - C^2 = \bar{A}^* X(I - P_A)X \bar{A} = R_A^* R_A, \quad R_A = (I - P_A)X \bar{A}.$$

Insert $I = P_A + (I - P_A)$ to prove the identity. It vanishes precisely on an X -invariant feature subspace. The actual empty shell $p = 37, a = 18, R = 35$ has two exterior states $u = 1, 12$ with $D = 7, 5$. On their constant line,

$$C = 6/35, \quad \bar{A}^* \Phi_3(X) \bar{A} = 0, \quad \Phi_3(C) = 17/343000 > 0.$$

The latter ordering would give a false positivity test. Its discarded quadratic relation norm is $1/1225$.

Scope and reproduction. The accompanying standard-library `verify.py` and `prime_certificate.json` reproduce (5)–(8), the full 9702-state census, all three hits and their original inverse data. Primality is checked by a recursive complete- $p-1$ order certificate (eight internal nodes, fifteen trial-division leaves). The bounded scan through $p \leq 1500$ checks 9531 shells and 20177 support choices without a false positive; it supplies no new prime coverage. The example (6) is not claimed least. Localizing matrices and Schur identities are classical; historical priority for this exact application is not asserted. Universal positivity in at least one shell of every prime remains unproved. All displayed sufficient implications have complete proofs above.

1. Elsholtz, C., & Tao, T. (2013). *J. Aust. Math. Soc.*, 94(1), 50–105. arXiv:1107.1010, §2, Propositions 2.2 and 2.6.
2. Lasserre, J.-B. (2011). *SIAM J. Optim.*, 21(3), 864–885. arXiv:1009.0125v3, (2.3)–(2.4), Theorem 3.2.
3. The Clankers (2026). *Joint Gamma Schur Return*, `JOINT_SCHUR_EXACT_INTEGRATION.tex`, JSA12–JSA20; Zeta repository commit `d1191f9c...ca7ff`.
4. The Clankers (2026). *Discrete-spectrum positivity and hard-shell continuations*, supplied 14 September package, `core.tex` and `second_hard_prime.json`. Exact pins accompany this source.